IEEE DataPort – MathWorks Challenge

IEEE SPS Bangalore Chapter

SPS Forum 2026

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Outline

- Challenge Problem Statement
- MATLAB & Simulink Use for Challenge
- LLM Use in Challenge Problem
- Requirements

IEEE DataPort

IEEE DataPort – an online platform for sharing, accessing, and managing datasets across technical domains.

- Supports upload, storage, discovery, and download of datasets.
- Increases data visibility, reuse, and citation.
- Supports benchmarking, validation, and reproducible research.
- Accelerates research and innovation across the technical community.
- Connects datasets to the broader IEEE ecosystem.



The Challenge

- **Goal:** Develop a robust and accurate AI models that can run on an edge device and classify motor imagery (MI) tasks from EEG (Electroencephalogram) recordings.
  - **Dataset:** Qing Zhou, "EEG Dataset of 7-day Motor Imagery BCI ", IEEE DataPort, November 2020, DOI:10.21227/f1c7-7x89 | Link on IEEE DataPort
- **Platform:** MATLAB or MATLAB with Python
  - Complimentary trials for MATLAB and associated tools provided to participating teams.
- **GenAI/LLM Use:** LLM/Copilot use allowed. Usage must be declared and prompts / logs shared for review.

****NOTE:** Participants accountable for submissions and should be able to explain code and answer questions on underlying logic and architecture.

BCI Dataset

- 20 subjects
- 7 sessions
- 6 runs / session
- 40 trials / run

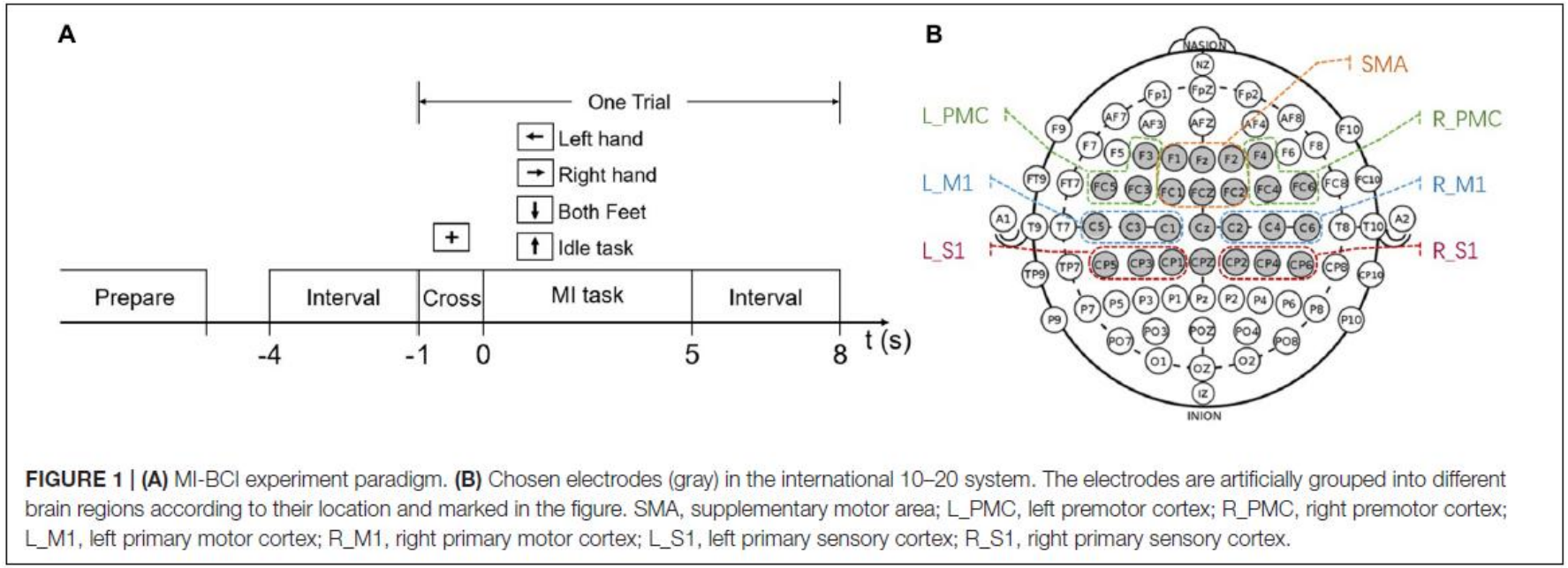


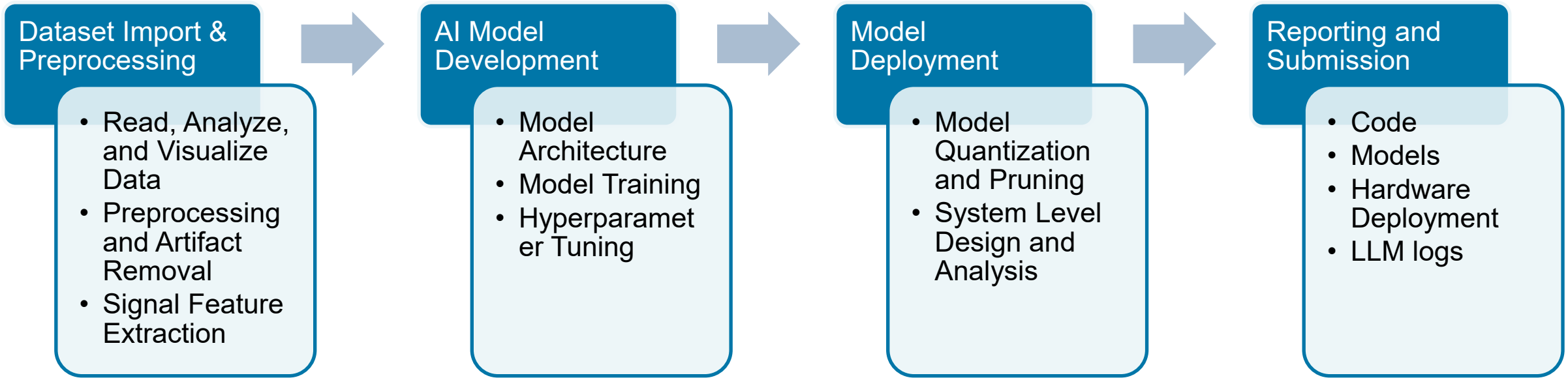
FIGURE 1 | (A) MI-BCI experiment paradigm. **(B)** Chosen electrodes (gray) in the international 10–20 system. The electrodes are artificially grouped into different brain regions according to their location and marked in the figure. SMA, supplementary motor area; L_PMC, left premotor cortex; R_PMC, right premotor cortex; L_M1, left primary motor cortex; R_M1, right primary motor cortex; L_S1, left primary sensory cortex; R_S1, right primary sensory cortex.

- For every run
  - 10 trials each for left hand, right hand, feet, idle

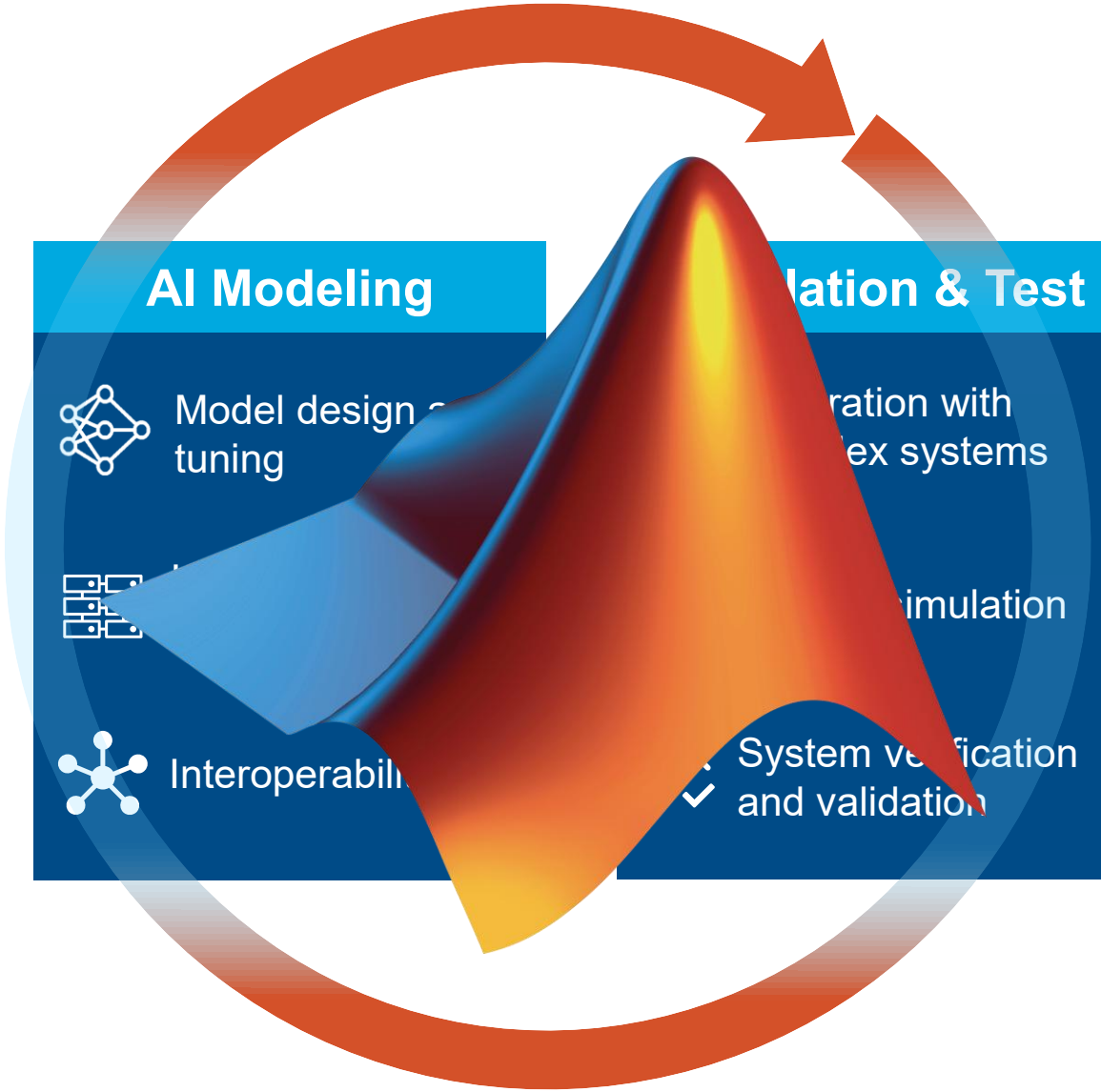
- Dataset has multi-day, multi-session recordings
  - Need to account for inter-subject and inter-session variability

What Participants Need to Build

- **Goal:** Develop a robust and accurate AI models that can run on an edge device and classify motor imagery (MI) tasks from EEG (Electroencephalogram) recordings.



AI-Driven System Design with MATLAB & Simulink



Data Preparation

- Data cleansing and preparation
- Human insight
- Simulation-generated data

AI Modeling

- Model design … tuning
- …
- Interoperabili…

…lation & Test

- …ration with …ex systems
- …imulation
- System ve…fication and validation

Deployment

- Embedded devices
- Enterprise systems
- Edge, cloud, desktop

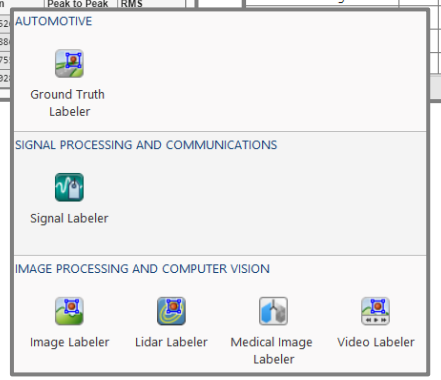
7

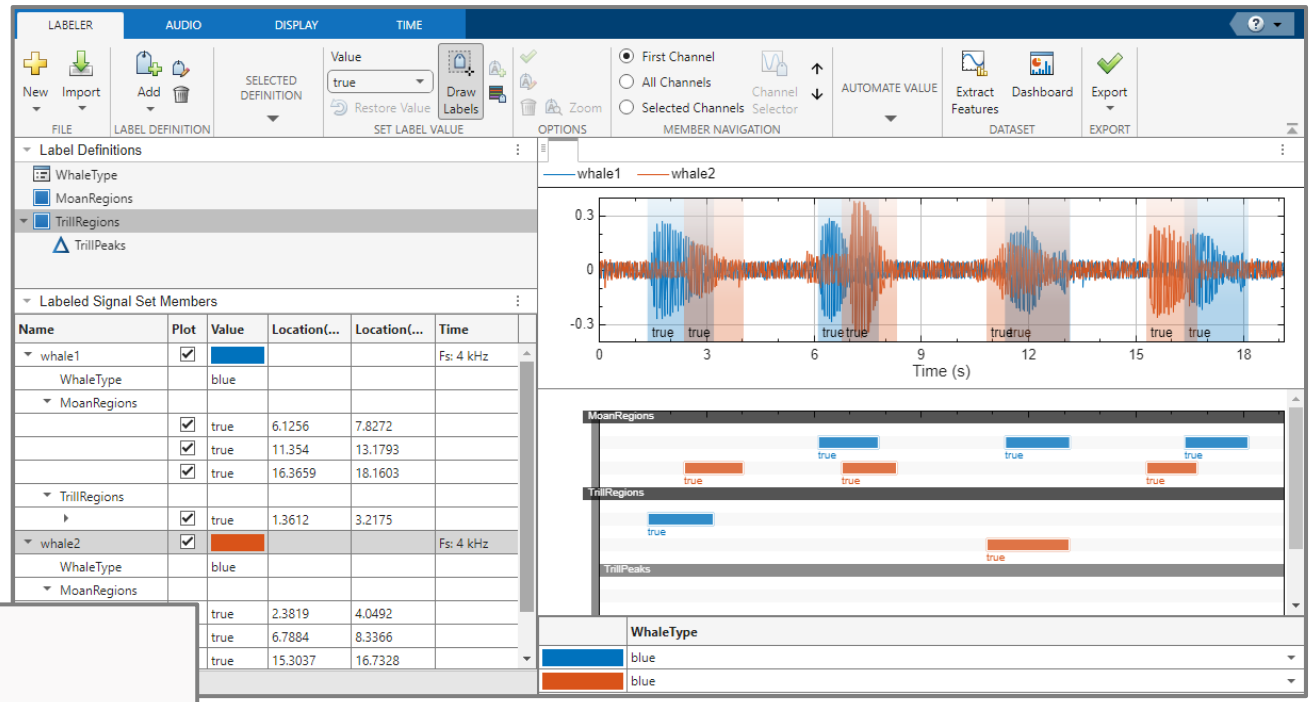
Spend less time preprocessing and labeling data

Synchronize disparate time series, filter noisy signals, automate signal labeling, and more



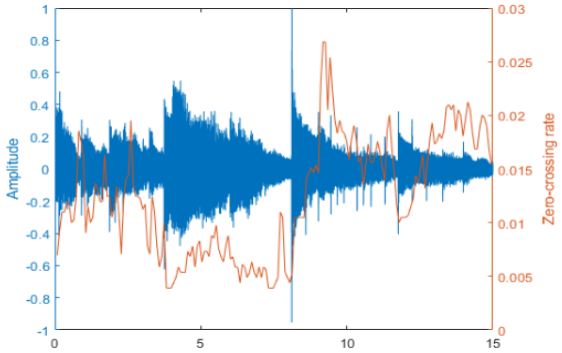
`signalAnalyzer`



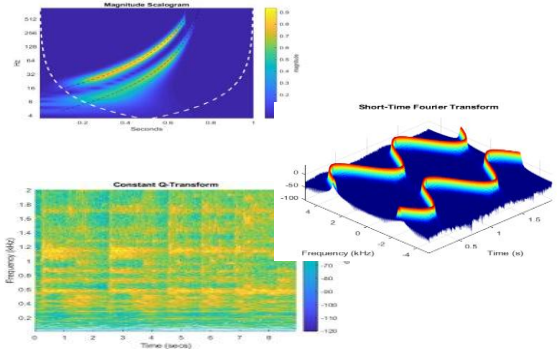


`signalLabeler`

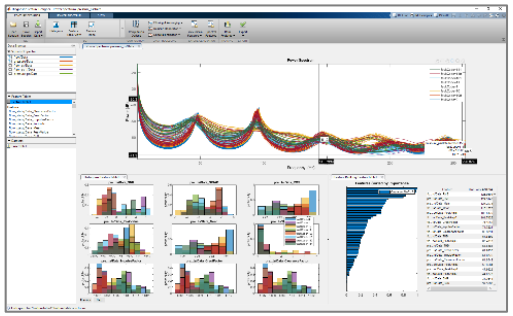
Transform raw data for useful modeling and analysis



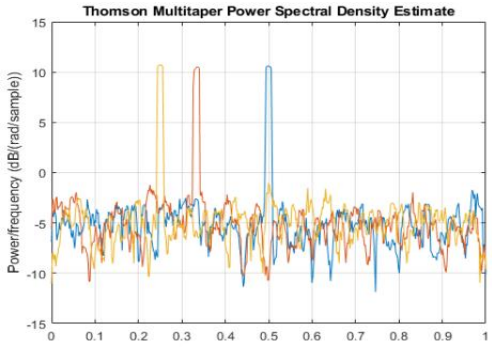
`signalTimeFeatureExtractor`



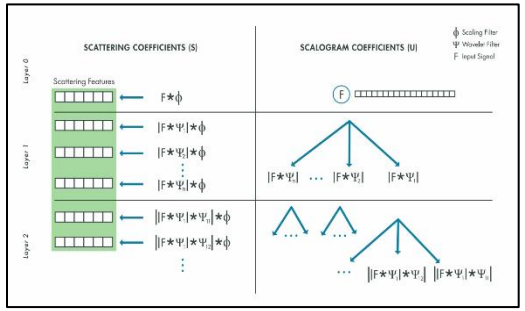
`signalFrequencyFeatureExtractor`



`diagnosticFeatureDesigner`



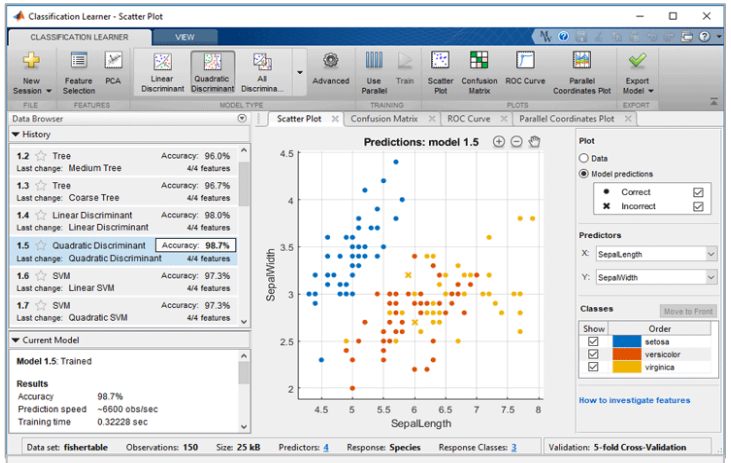
`signalTimeFrequencyFeatureExtractor`



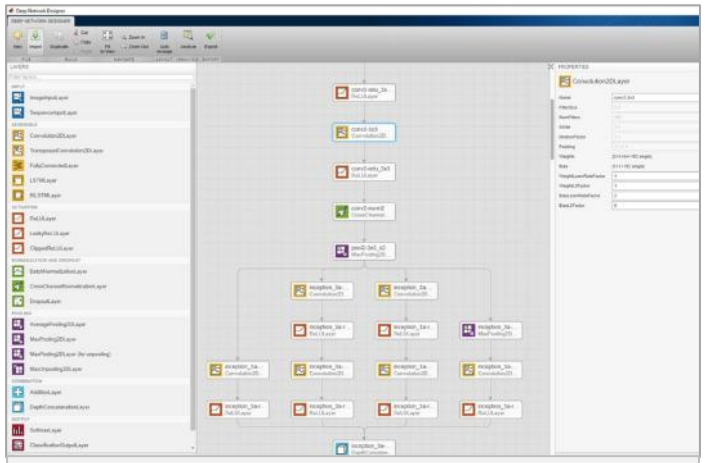
`Wavelet based features`

Increase productivity using apps for design and analysis

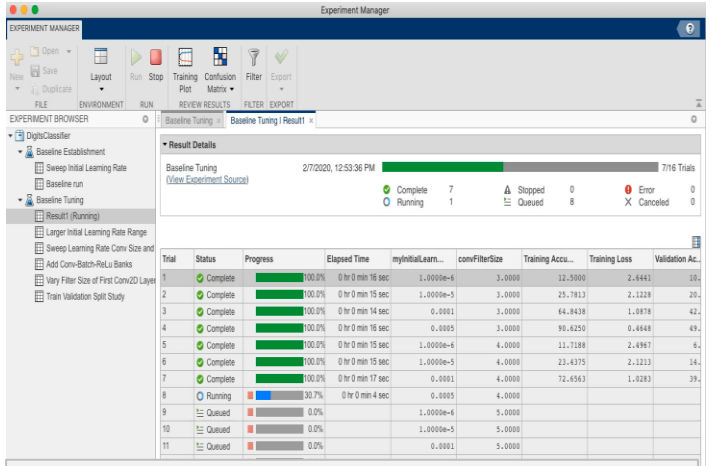
Use MATLAB Apps to explore a range of classifiers, train regression models, design deep learning networks, tune hyperparameters, and more.



Learner apps to try different algorithms and find the best fit for data sets.



Deep Network Designer app to build, visualize, and edit deep learning networks.

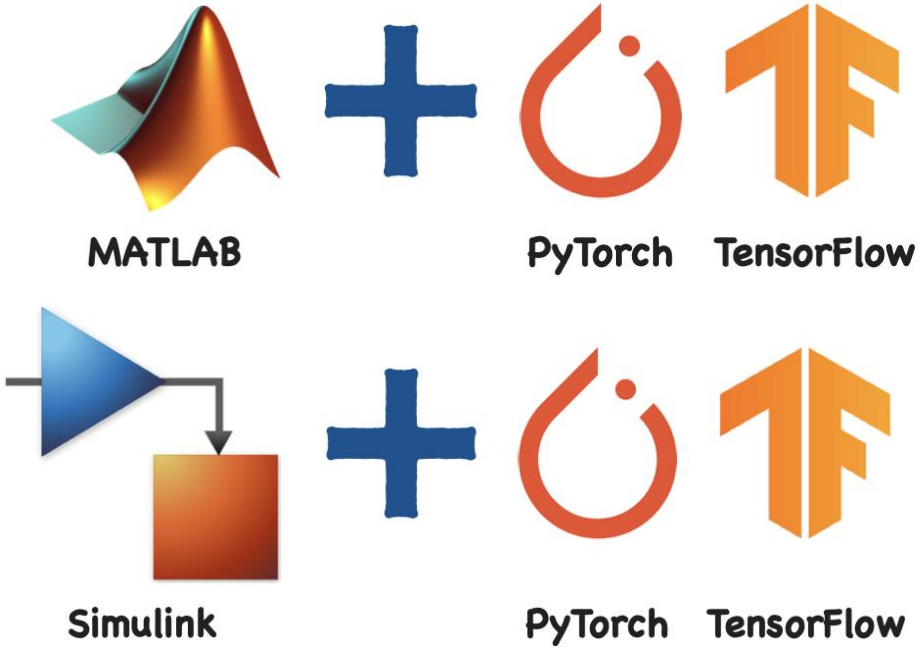


Experiment Manager app to run AI experiments to train models and compare results.

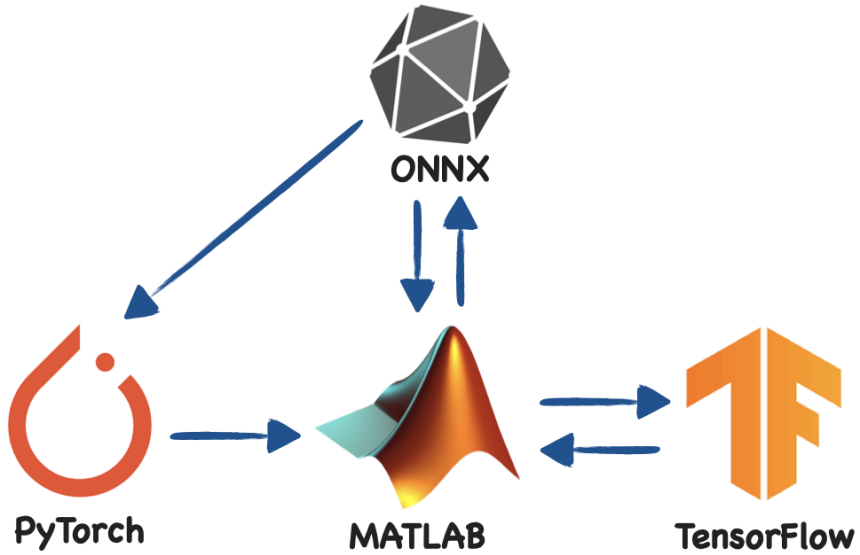
Combine MATLAB with Python for AI

Use MATLAB with Python for AI Modeling

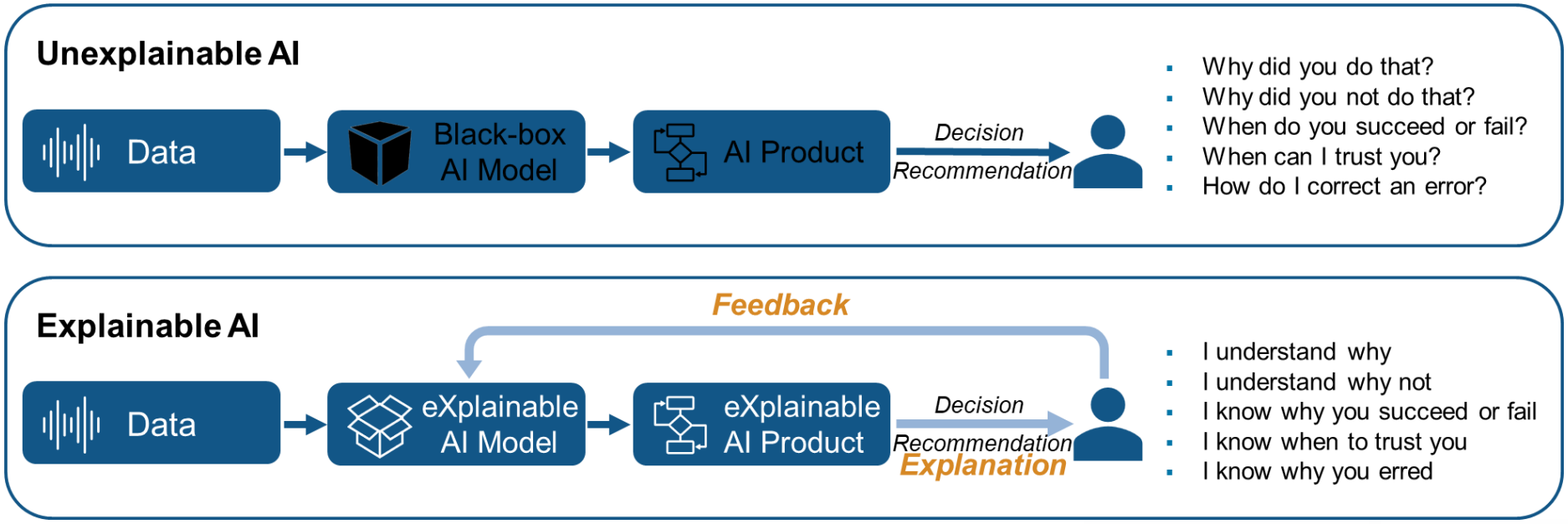
Co-execute

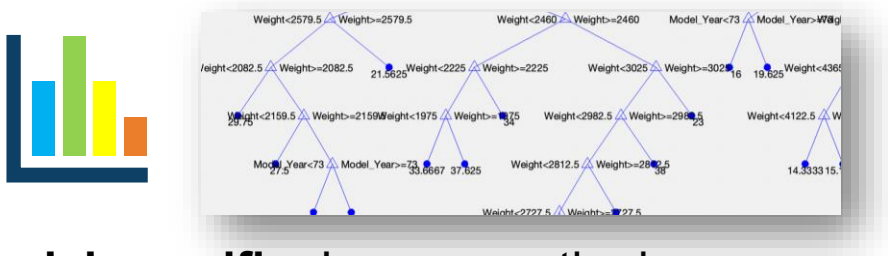


Exchange Models

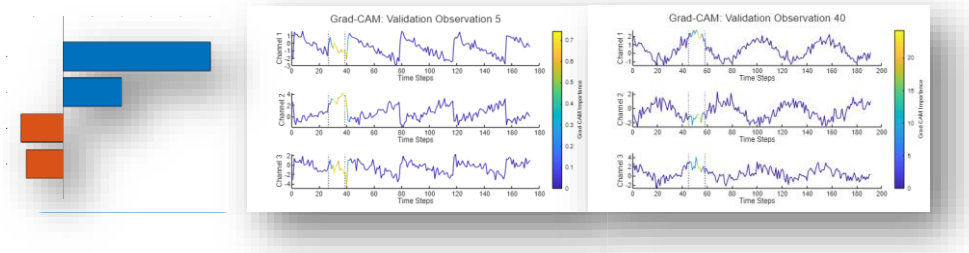


Explainable AI facilitates informed decisions





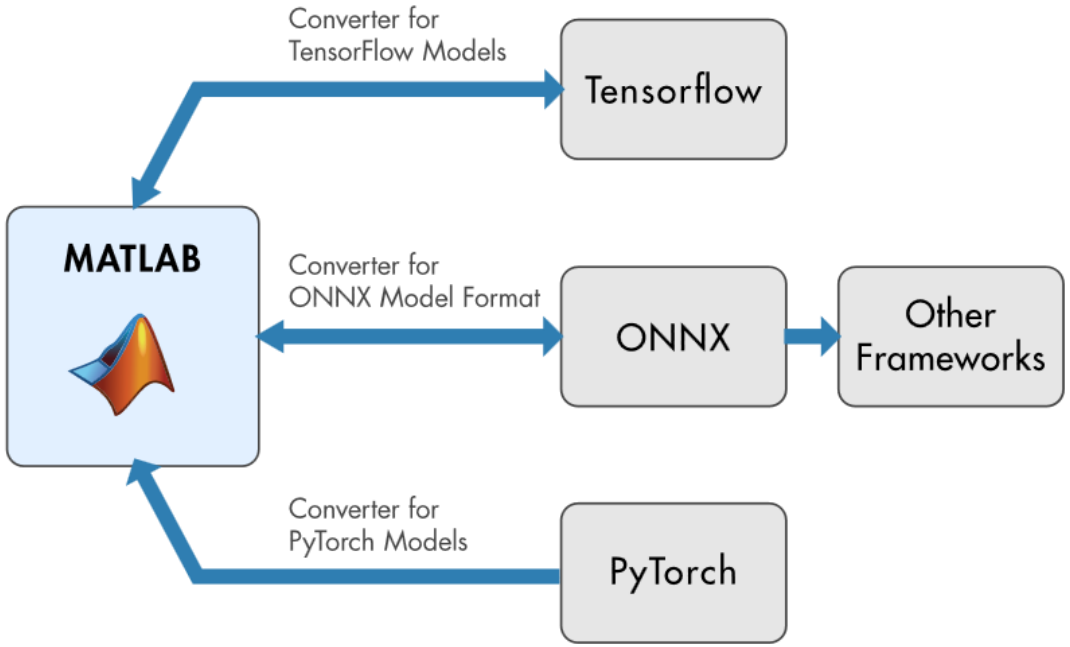
Model-specific: Leverages the inner workings of a model to interpret its result

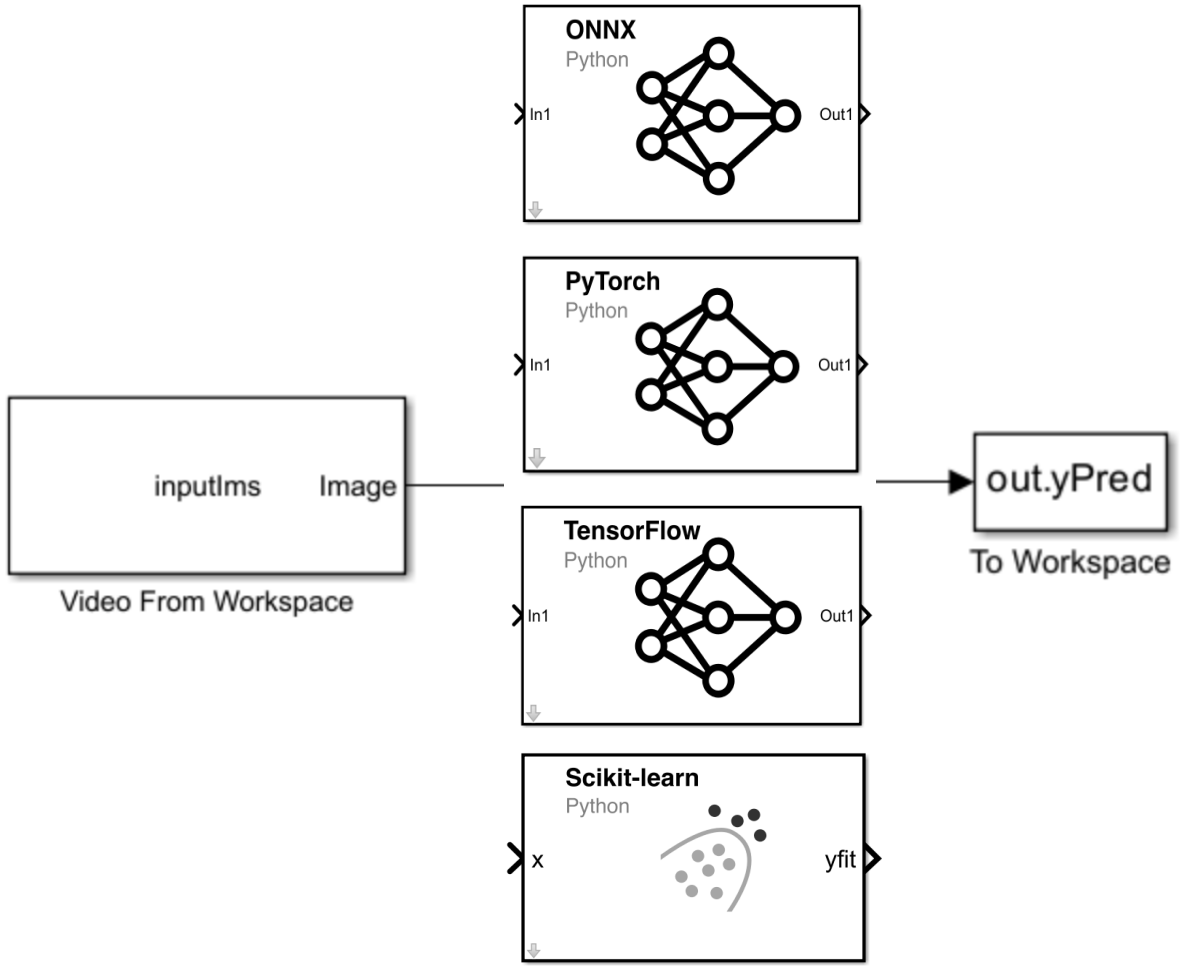


Model-agnostic: Analyzes the input features in relation to the output

Integrate MATLAB, Simulink and Python frameworks for AI

Share MATLAB and Python-based deep learning models, directly use Python-based AI models for Simulation in Simulink.



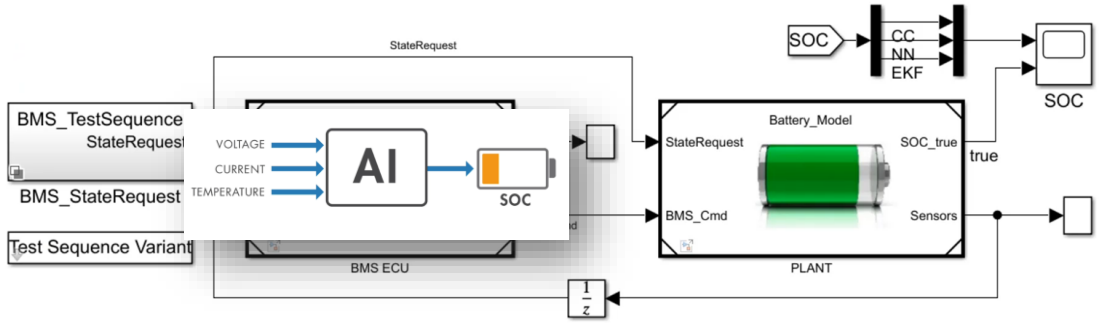


AI models drive advancements in engineered systems

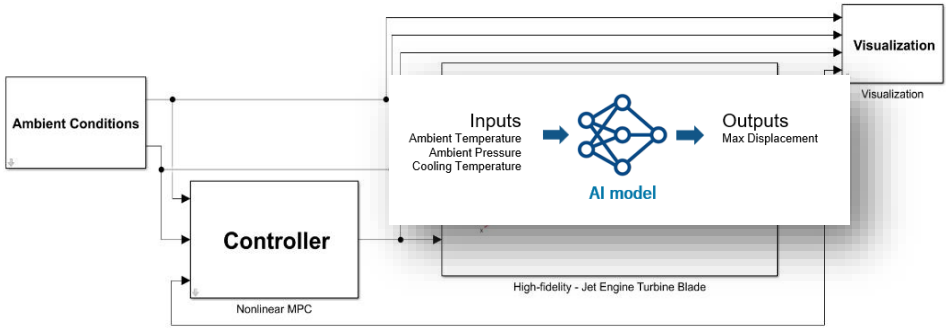
Adding data-driven components using AI can allow new capabilities, reduce costs, increase predictive accuracy, and allow earlier, more robust system testing



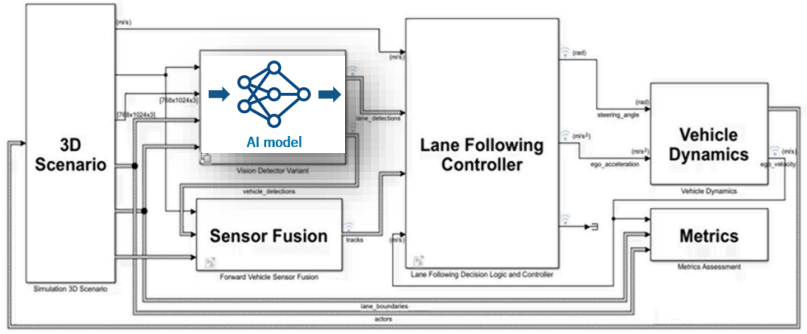
In ***Battery Management System:***
Virtual sensor for battery state of charge (SOC) estimation



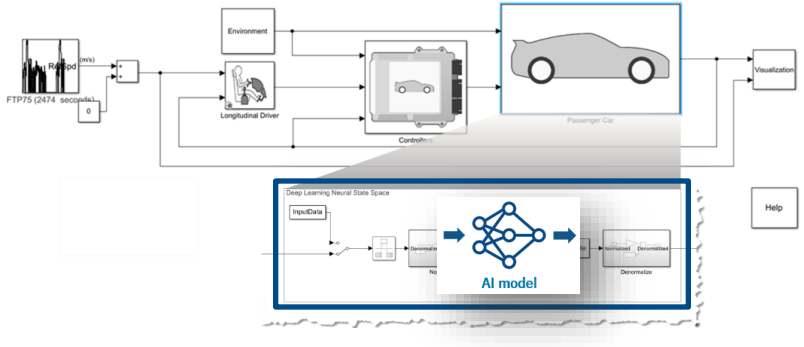
In ***Jet Engine Closed-Loop Control:***
Reduced-order model (ROM) of blade structural behavior (FEA model)



In ***Highway Lane Following System:***
Computer vision model detects lanes for controller



In ***Vehicle Drive-Cycle Virtual Test Bench:***
ROM replaces the vehicle for rapid design and testing of control systems

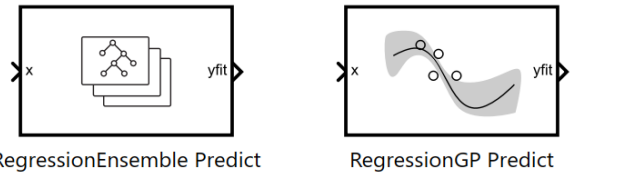


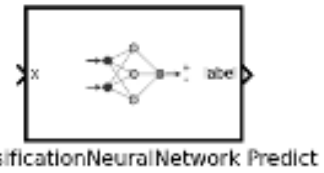
Data Preparation | AI Modeling | **Simulation & Test** | Deployment

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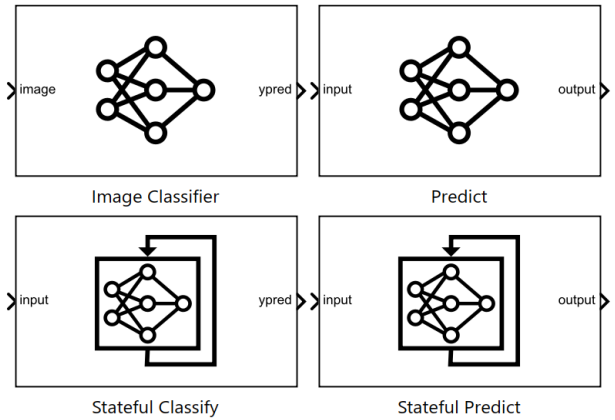
Simulink includes more AI blocks for more applications

AI core

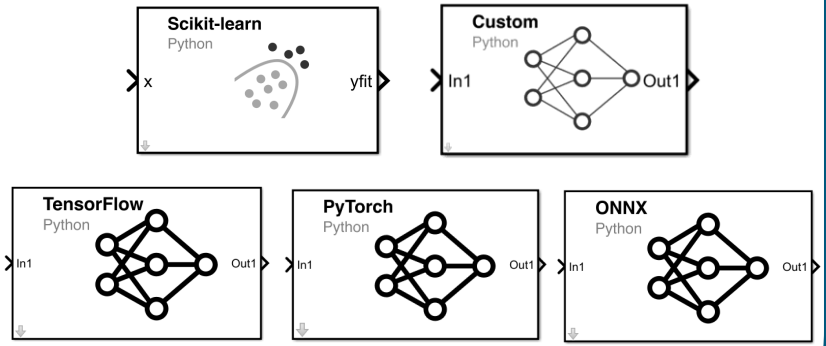




Regression and classification blocks
(Statistics and Machine Learning Toolbox)

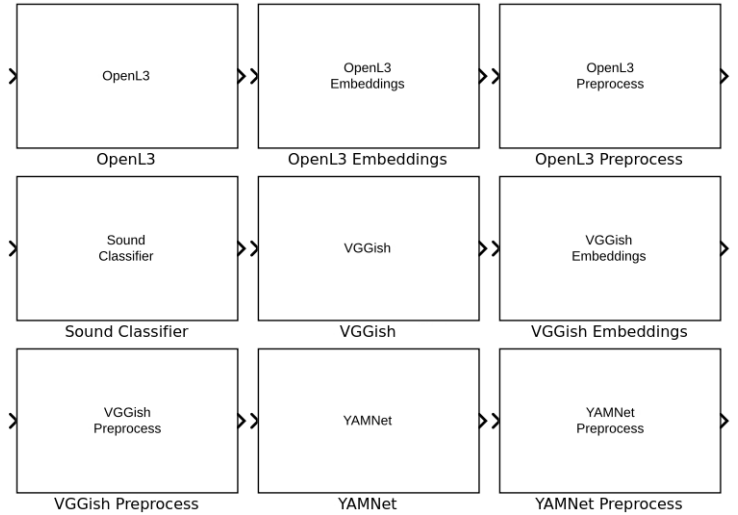


Neural Network blocks
(Deep Learning Toolbox)

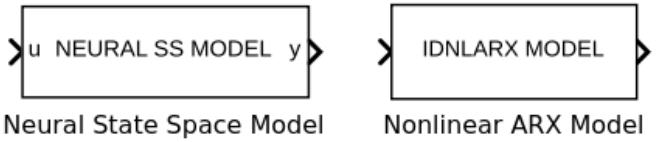


Co-execution blocks

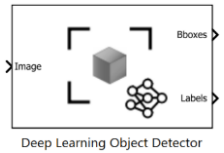
Specialized



Audio Toolbox

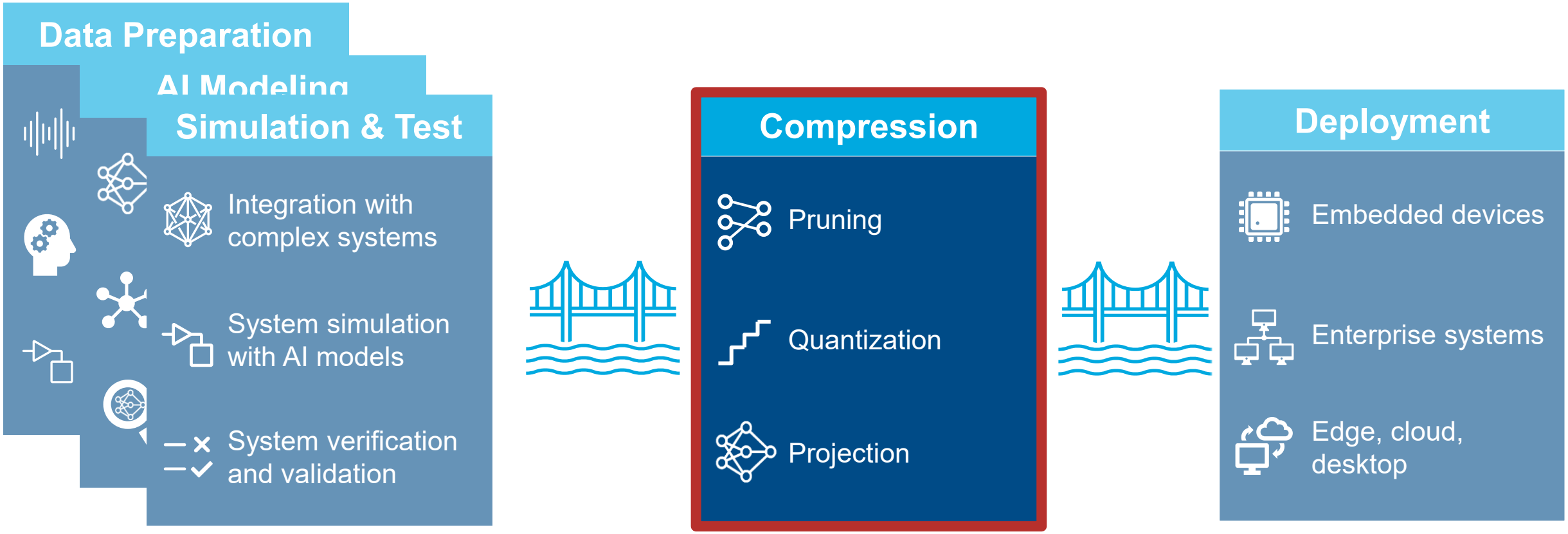


System Identification Toolbox



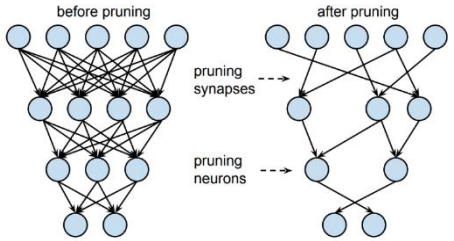
Computer Vision Toolbox

Model compression bridges the gap between design & deployment

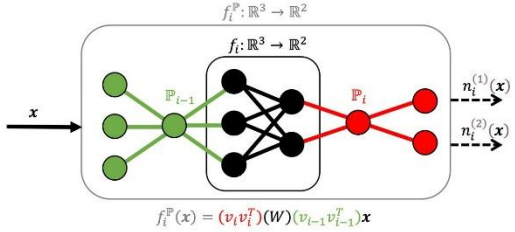


Model Compression

Structural Compression

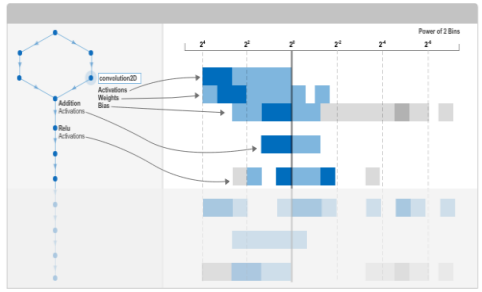


Pruning convolutional neural networks



Projection of deep neural networks

Datatype Compression

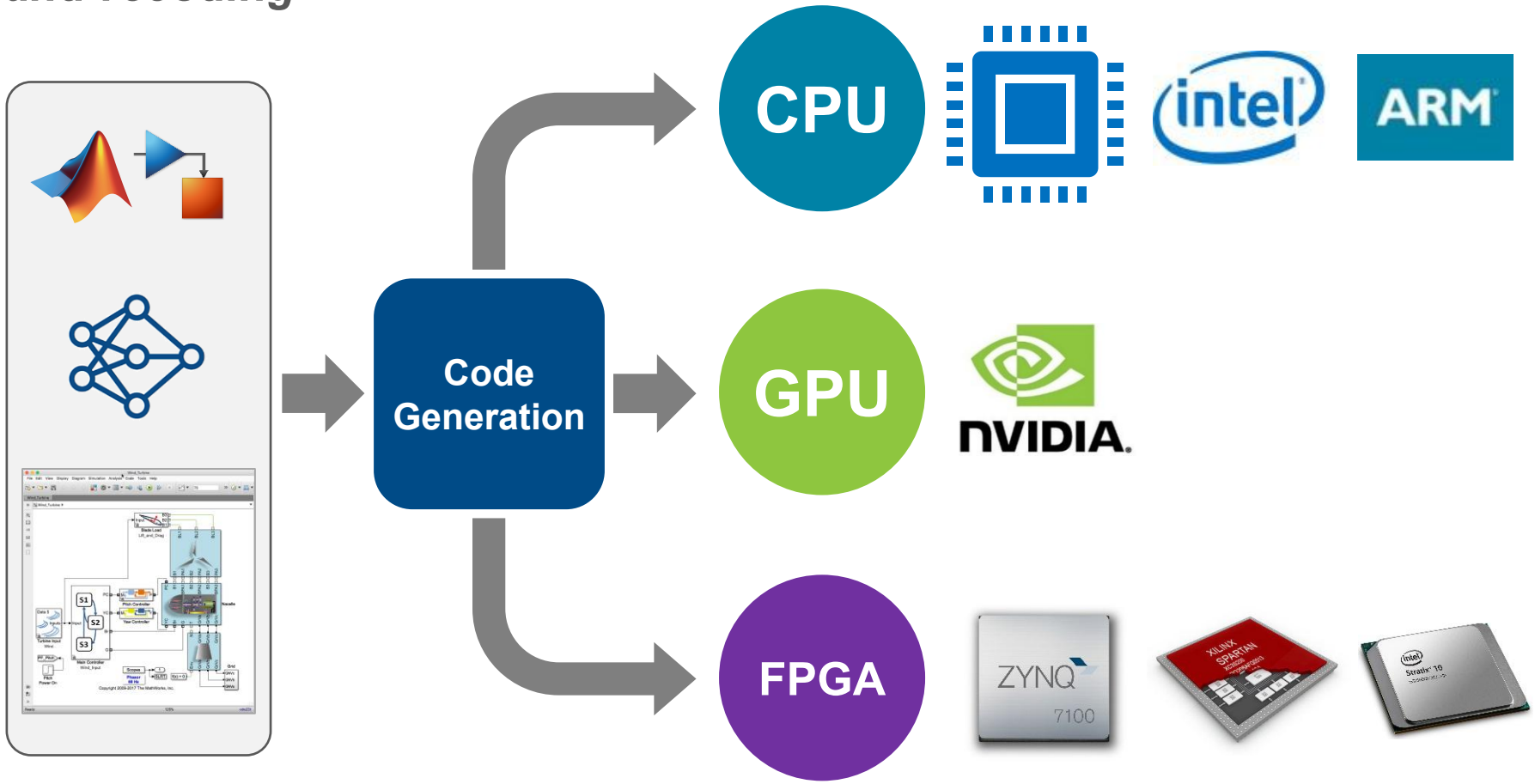


Quantization of network weights to lower precision datatypes (bfloat16, int8)

[Quantization, Projection, and Pruning]

Automatically generate code for embedded Devices

Test trained models on low-cost prototyping boards or take models to production *without hand-recoding*



Get Started with AI in MATLAB and Simulink

Start directly in the browser with no download and installation



Computer Vision for Engineering and Science



Deep Learning for Computer Vision



Image Processing for Engineering and Science



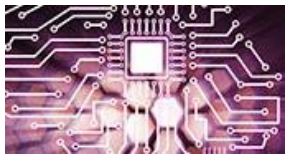
Practical Data Science with MATLAB

coursera

Specializations – multiple-course series



Machine Learning Onramp



Machine Learning with MATLAB



Deep Learning Onramp



Deep Learning with MATLAB



Reinforcement Learning Onramp

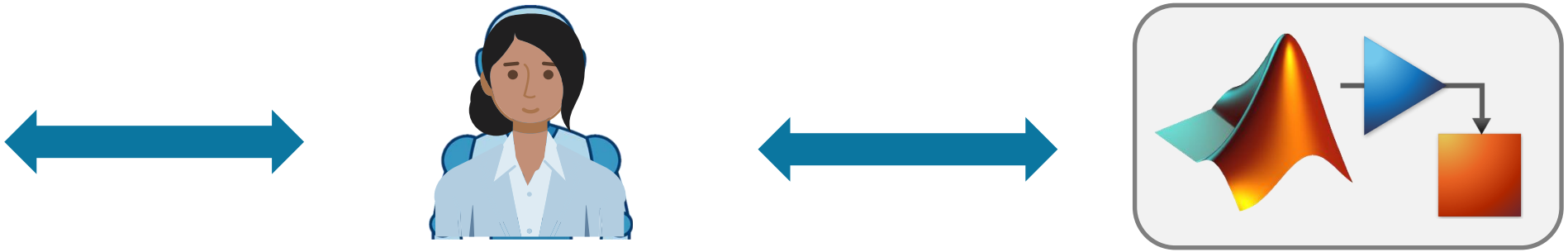


Computer Vision Onramp

https://matlabacademy.mathworks.com/#ai

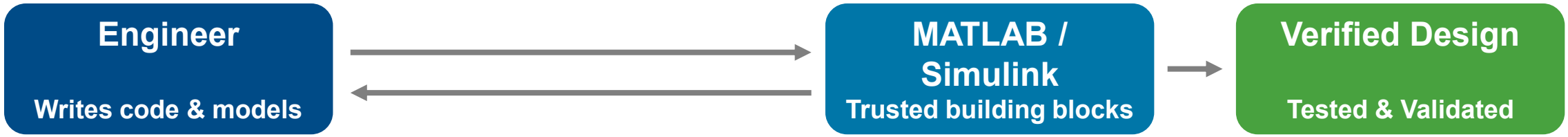
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Generative AI is transforming engineering

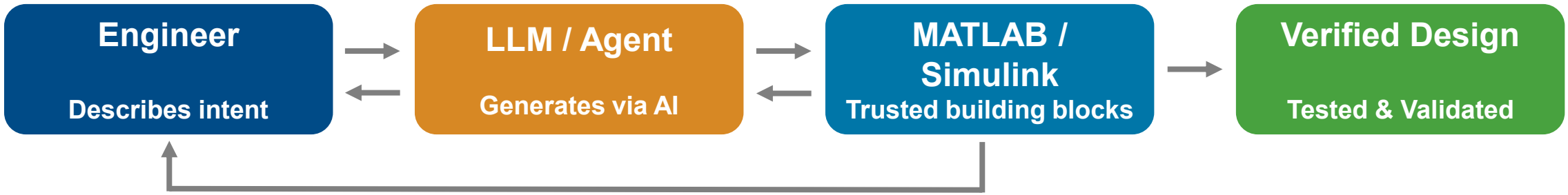


GenAI changes how you create - not engineering accountability

Today



With AI



What doesn't change when you add AI

The engineer owns the design. The same engineering evidence verifies it.

Accelerate your MATLAB and Simulink productivity with GenAI

Agentic Workflows



External GenAI Agents

use MATLAB or Simulink with external Agents via MCP or VS Code



Build Agents in MATLAB

call LLMs from MATLAB code

AI Assistants



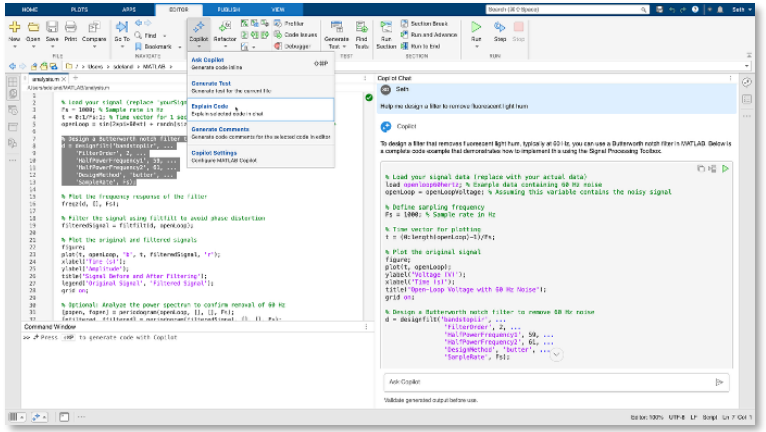
MATLAB and Simulink Copilots

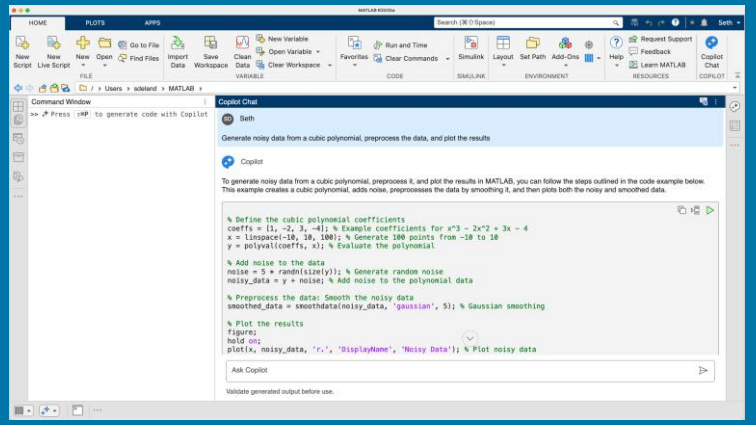
optimized for MATLAB and Simulink

MATLAB Copilot

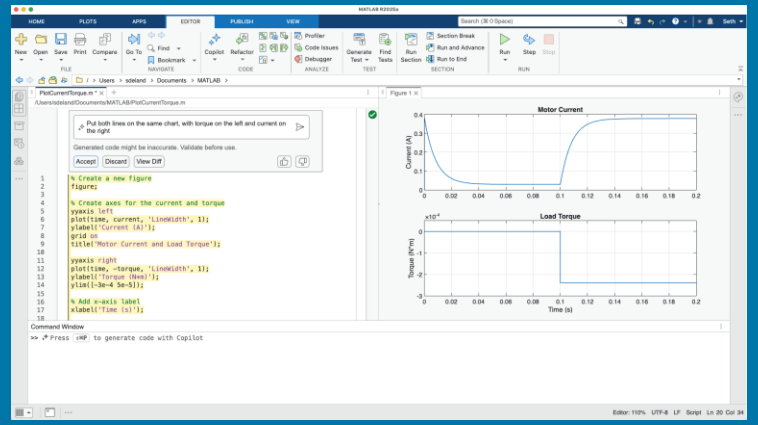
R2025a

- Customized for MATLAB and Responses are based on the latest MathWorks documentation
- Integrated into MATLAB and tuned for engineering and scientific workflows

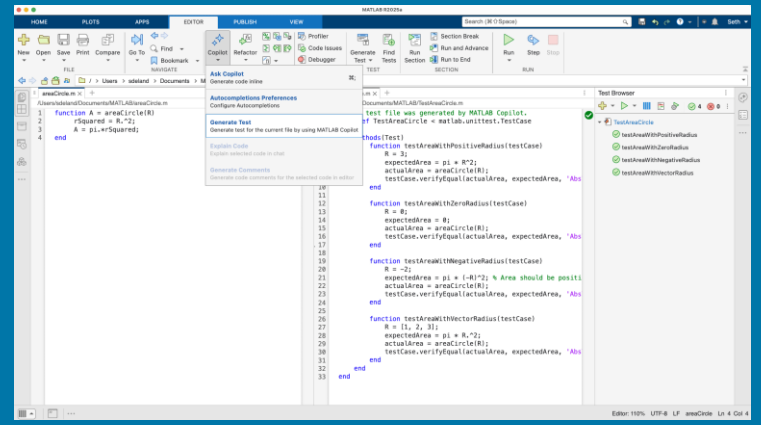




Copilot Chat



Copilot in the Editor and Command Window

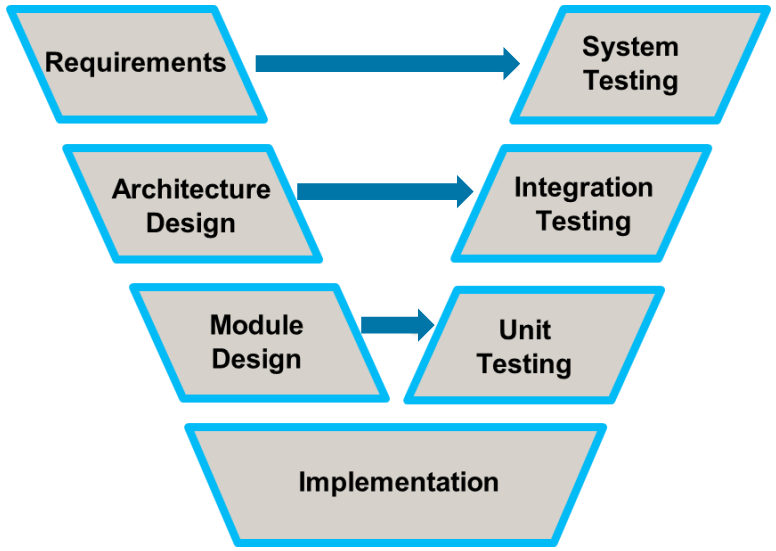


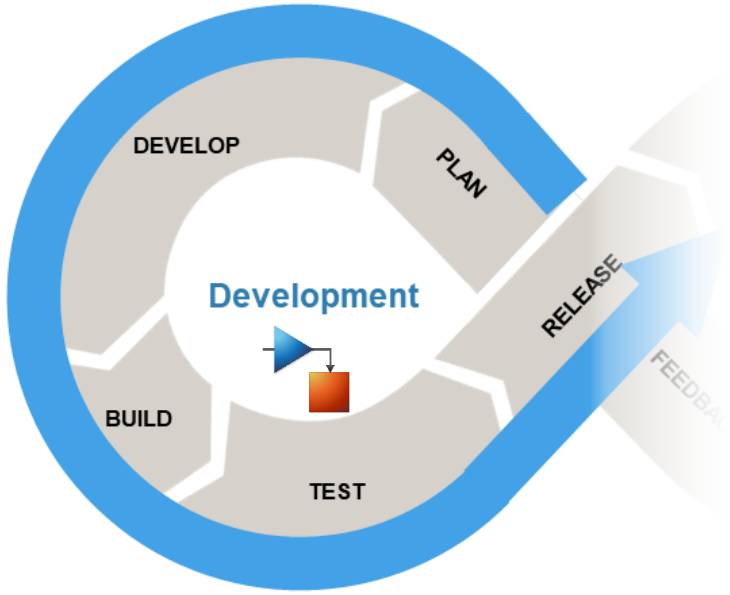
Copilot Actions

Simulink Copilot

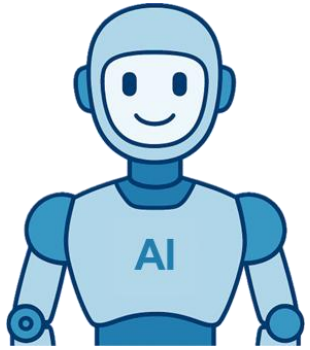
R2026a

An AI assistant optimized for Simulink and designed to improve your productivity with Model-Based Design



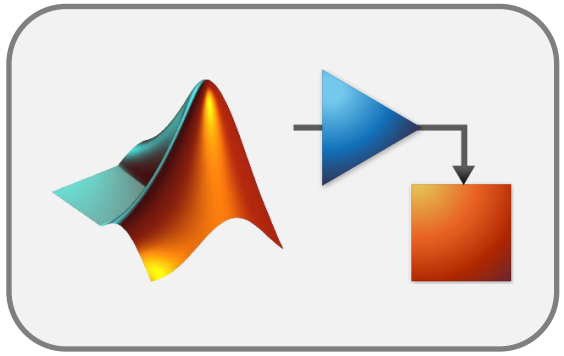


Agentic workflows are enabled by Model Context Protocol (MCP)



Cloud providers
OR
Local models





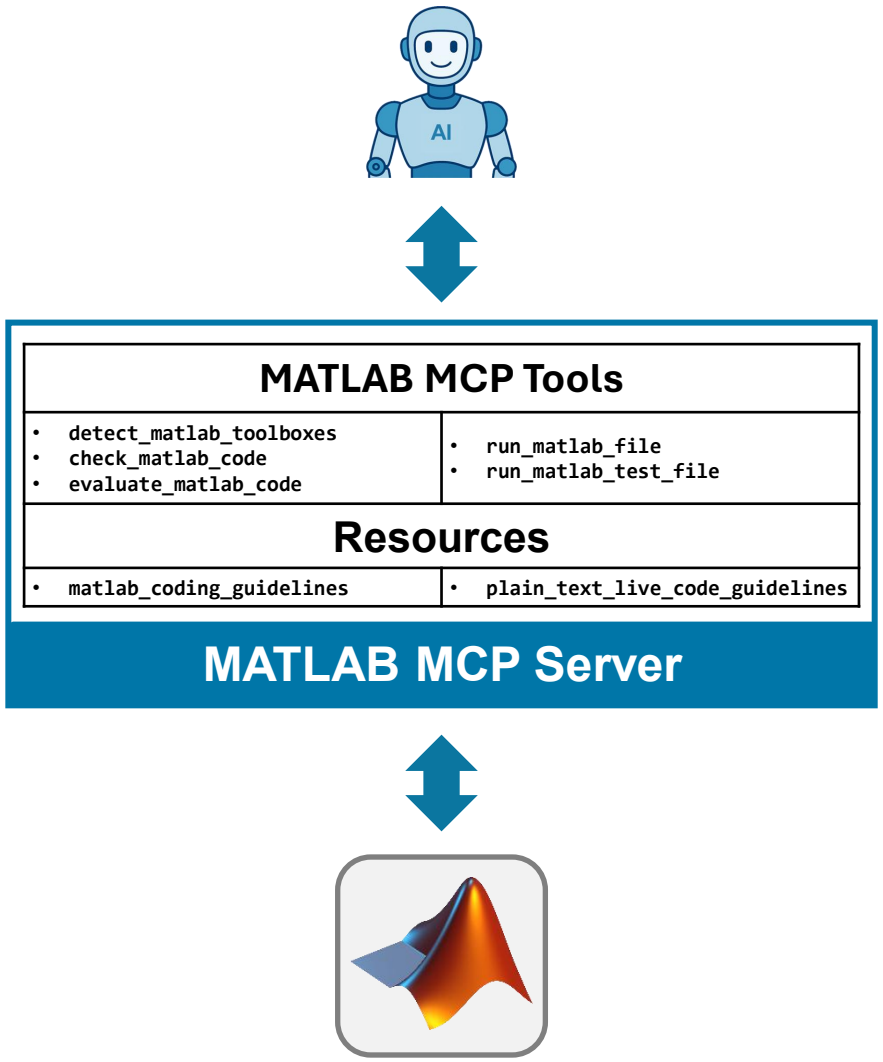
Tools

MATLAB MCP Server

Enabling GenAI Agentic Workflows with MATLAB

A **free, open-source** local MCP server that connects GenAI assistants to a **locally** licensed MATLAB, enabling intelligent automation and code execution

- No additional license required – just connect and go
- Preserves privacy with local execution
- Start a new MATLAB session or connect to an existing session



https://github.com/matlab/matlab-mcp-server

How can we improve agent behavior?

Skills transform prompts into reliable, domain-aware, reusable workflows

- Right knowledge at the right time
- Reliability and output quality
  - Encoding proven procedures
  - Grounding against hallucination
- Knowledge the model may not have
  - Domain and proprietary expertise
  - Institutional memory

/skills/skill-name/SKILL.md

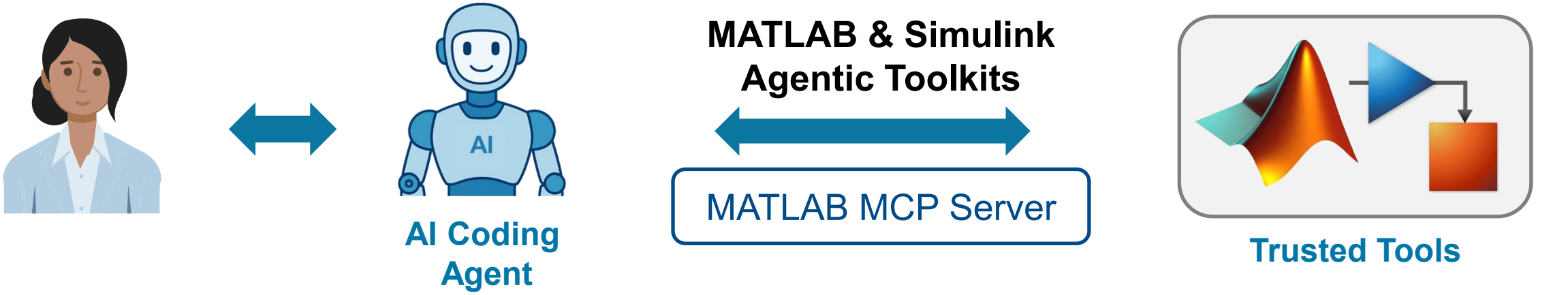
```
---
name: skill-name
description: Defines the conditions under which
the skill should or should not be triggered
---


# Instructions
This section is loaded by the agent only if it
determines, based on the skill description, that
the content may be relevant to the current task.
```

Skills tells your agent how to do things correctly the first time around

MATLAB & Simulink Agentic Toolkits

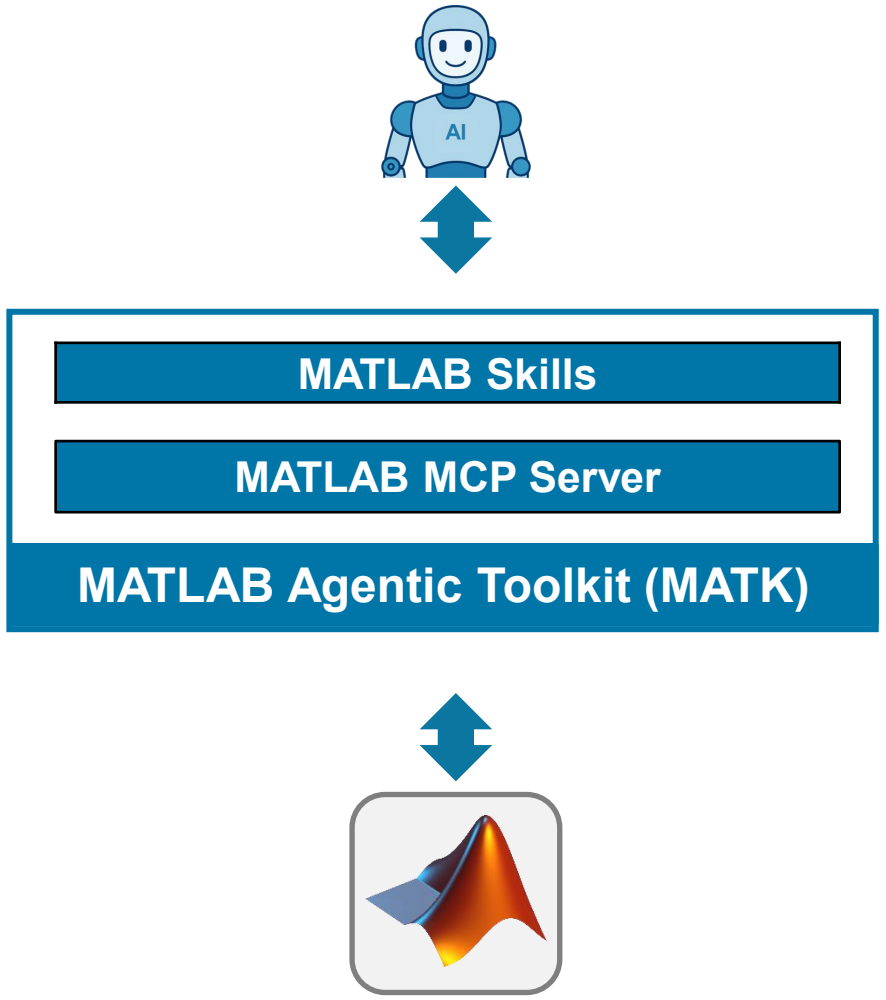
Bring Trusted Capabilities to AI Agents



MATLAB Agentic Toolkit

Connect trusted MATLAB capabilities to AI agents to make engineering and scientific workflows agent-ready

- MATLAB MCP Server
- Expert Skills Catalog encode MATLAB best practices, guiding AI in testing, debugging, and code modernization.
  - MATLAB Software Development
  - Signal Processing
  - Image Processing and Computer Vision
  - Wireless Communications
  - .. and many more!!



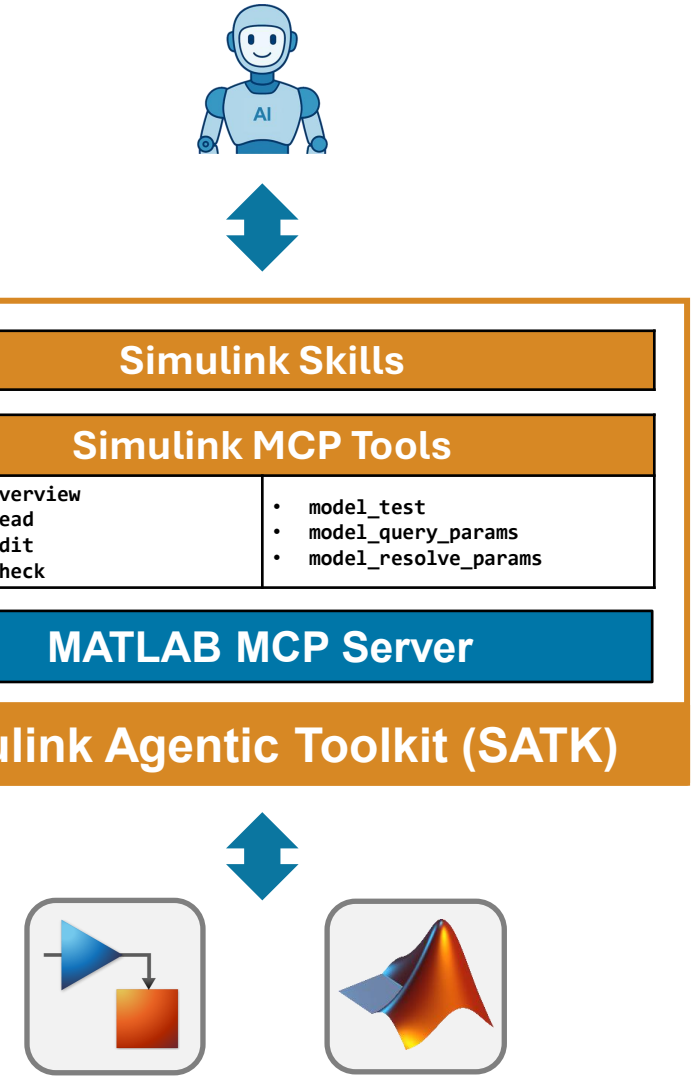
https://github.com/matlab/matlab-agentic-toolkit

Simulink Agentic Toolkit

Connect agents to Simulink giving them the ability and the knowledge to work with Simulink models effectively.

- Builds on top of the MATLAB MCP Server
- MCP tools optimized for reading, editing, querying, testing, and checking Simulink models
- Agent skills encoding Model Based Design best practices for model building, plant specification, testing, requirements, and architecture modeling

https://github.com/matlab/simulink-agentic-toolkit



IEEE DataPort – MathWorks Challenge

- **Goal:** Develop a robust and accurate AI models that can run on an edge device and classify motor imagery (MI) tasks from EEG (Electroencephalogram) recordings.
  - **Dataset:** Qing Zhou, "EEG Dataset of 7-day Motor Imagery BCI ", IEEE DataPort, November 2020, DOI:10.21227/f1c7-7x89 | Link on IEEE DataPort
- **Platform:** MATLAB or MATLAB with Python
  - Complimentary trials for MATLAB and associated tools provided to participating teams.
- **GenAI/LLM Use:** LLM/Copilot use allowed. Usage must be declared and prompts / logs shared for review.

****NOTE:** Participants accountable for submissions and should be able to explain code and answer questions on underlying logic and architecture.

IEEE DataPort – MathWorks Challenge

Submission Requirements

- **Code:** Complete code including MATLAB scripts/LiveScripts and Python notebooks (if any) used for preprocessing, model training, model tuning, model quantization, and prediction. Code should be well-commented and reproducible.
- **Model:** Trained model file(s) in a common format (e.g., ONNX, TensorFlow SavedModel, PyTorch state_dict, MATLAB .mat, pickle).
- **Prediction File:** A CSV file with predictions on the provided test set, formatted as specified.
- **Technical Report (Max. 2 pages):** Methodology, feature engineering, model architecture, handling of cross-session variability, challenges faced, and insights gained.
- **Presentation**

IEEE DataPort – MathWorks Challenge

- **Goal:** Develop a robust and accurate AI models that can run on an edge device and classify motor imagery (MI) tasks from EEG (Electroencephalogram) recordings.

- **Prizes:** Cash Prizes for Top 3 Finalists, Certificates from IEEE SPS and MathWorks, Feature on IEEE SPS Website and Social Media

Challenge Page:

